

Health Intelligence Report

Patient: Rachel Bertler
Generated: May 29, 2026

NOTICE: This report identifies patterns in health data to support informed conversations with a healthcare provider. It is not a medical diagnosis. Always consult a qualified clinician.

SUMMARY

This 30s female patient presents with complex neurodevelopmental and mental health diagnoses (ADHD, provisional ASD, GAD, MDD) alongside metabolic optimization efforts (Wegovy for weight management) and a notable GI history. Cross-domain analysis reveals several patterns worth clinical discussion: persistently deficient vitamin D despite 5+ years of documented low levels, a microbiome profile suggesting dysbiosis with potential *H. pylori* colonization and occult GI bleeding, and declining testosterone alongside chronic stress and burnout. Medication adjustments show sensitivity to side effects across multiple drug classes.

PATTERNS IDENTIFIED

Chronic Vitamin D Deficiency Despite Known Low Levels Since 2020

Confidence: Strong pattern

RELATED CONDITIONS

Major Depressive Disorder · Generalized Anxiety Disorder · Raynaud's Phenomenon

SUPPORTING EVIDENCE

- Vitamin D 10 ng/mL (severely deficient) — Lab Results, March 2020
- Vitamin D 14.8 ng/mL (still deficient, reference 30–100 ng/mL) — Lab Results, July 2025
- Raynaud's Phenomenon without gangrene — Health History, April 2026
- Major Depressive Disorder, Generalized Anxiety Disorder — Health History, April 2026

GI Dysbiosis Pattern with *H. pylori*, Occult Bleeding, and Depleted Beneficial Bacteria

Confidence: Strong pattern

RELATED CONDITIONS

Pelvic Pain · Endometriosis · Major Depressive Disorder · ADHD

SUPPORTING EVIDENCE

- Helicobacter pylori* 1420 genome equivalents/g stool (flagged positive) — Lab Results, May 2023
- Occult Blood (FIT) 67 ug/g (flagged abnormal) — Lab Results, May 2023
- Faecalibacterium prausnitzii* 888 genome equivalents/g (ref: 1,000–500,000,000) — severely depleted — Lab Results, May 2023
- Escherichia* spp. 1,050,000 genome equivalents/g (ref: 3,700,000–3,800,000,000) — markedly low — Lab Results, May 2023
- Secretory IgA 239 ug/g (ref: 510–2010) — flagged low — Lab Results, May 2023
- Anti-gliadin IgA 500 U/L (flagged elevated) — Lab Results, May 2023
- History of pelvic pain and endometriosis — Health History, August 2007

Declining Testosterone in Context of Chronic Stress and Burnout

Confidence: Possible pattern

RELATED CONDITIONS

Burnout · Work Stress · Major Depressive Disorder

SUPPORTING EVIDENCE

- Total testosterone 60 ng/dL (May 2020, flagged elevated for female range 8–60) — Lab Results, May 2020
- Total testosterone 34 ng/dL (within normal range but 43% decline) — Lab Results, July 2025
- Occupational burnout, chronic work stress, considering career change — Health History, April 2026
- Major Depressive Disorder, current episode — Health History, April 2026
- Weight loss trend: 143.8 lbs ! 134 lbs over 1 month on Wegovy — Vitals, April–May 2026

Low-Normal CO2 Trend Suggesting Possible Subclinical Metabolic Acidosis

Confidence: Possible pattern

RELATED CONDITIONS

Generalized Anxiety Disorder · ADHD

SUPPORTING EVIDENCE

- Total CO2 20 mmol/L (at lower limit of normal, ref 20–29) — Lab Results, March 2020
- Total CO2 19 mmol/L (flagged below reference range 20–29) — Lab Results, July 2025
- Lisdexamfetamine (stimulant) active since April 2026 — Health History, April 2026

Medication Sensitivity Pattern Across Multiple Drug Classes

Confidence: Strong pattern

RELATED CONDITIONS

Autism Spectrum Disorder · ADHD · Generalized Anxiety Disorder · Major Depressive Disorder

SUPPORTING EVIDENCE

- Guanfacine discontinued due to paradoxical reaction — Health History, April–May 2026
- Methylphenidate discontinued due to ineffective, returned to Vyvanse — Health History, April–May 2026
- Wegovy 0.5 mg discontinued due to fullness and lack of hunger side effects, dose reduced to 0.25 mg — Health History, April–May 2026
- Mirtazapine discontinued to switch to Zoloft — Health History, April–May 2026
- Provisional Autism Spectrum Disorder diagnosis (evaluation pending) — Health History, April 2026

Borderline Low Ferritin Trend Despite Normal Hemoglobin

Confidence: Possible pattern

RELATED CONDITIONS

Endometriosis · Pelvic Pain · ADHD

SUPPORTING EVIDENCE

- Ferritin 16 ng/mL (ref 15–150, just above lower limit) — Lab Results, March 2020
- Ferritin 21 ng/mL (minimal improvement over 5 years) — Lab Results, July 2025
- Iron saturation declined from 32% ! 20% — Lab Results, March 2020 to July 2025
- History of endometriosis and pelvic pain (potential chronic blood loss) — Health History, August 2007
- Occult blood detected in stool — Lab Results, May 2023

INFORMATION GAPS

- Follow-up H. pylori testing post-2023 detection to confirm eradication or persistence
- Vitamin D supplementation dose and adherence records to explain persistent deficiency over 5+ years
- Colonoscopy or upper endoscopy to investigate occult GI bleeding detected in May 2023
- Repeat comprehensive stool analysis to assess whether microbiome abnormalities (depleted F. prausnitzii, low secretory IgA) have persisted or resolved
- Cortisol testing (morning, evening, or 24-hour urine) to assess HPA axis function given documented burnout and declining testosterone
- Formal ASD diagnostic evaluation results (provisional diagnosis from April 2026)
- Celiac serology (tissue transglutaminase IgA, total IgA) to follow up on elevated anti-gliadin IgA and GI symptoms
- Arterial blood gas or anion gap calculation to further characterise low CO2 pattern
- Iron studies trend monitoring and menstrual history to contextualise borderline ferritin and declining saturation

TALKING POINTS FOR YOUR PROVIDER

- Why has vitamin D remained deficient for 5+ years — is supplementation being prescribed and monitored? Given the associations between vitamin D deficiency and both mood disorders and autoimmune conditions (Raynaud's), should a structured repletion protocol with follow-up testing be established?
- The 2023 stool panel showed H. pylori colonization, occult bleeding, severely depleted keystone bacteria (F. prausnitzii), low mucosal immunity (secretory IgA), and elevated anti-gliadin antibodies. Was H. pylori treated? Should repeat testing be done given the occult bleeding? Could GI inflammation be contributing to fatigue or mood symptoms given the gut-brain axis relevance to ADHD/depression?
- Testosterone has dropped significantly (60 ! 34 ng/dL) during a period of documented chronic stress, burnout, weight loss, and depression. Could HPA axis dysregulation or energy deficit from rapid weight loss be contributing? Is this decline clinically

relevant to energy, motivation, or mood symptoms?

- CO₂ has been at or below the lower limit of normal for 5+ years (20 !' 19 mmol/L). Could this reflect compensatory hyperventilation (anxiety-related), renal bicarbonate wasting, or dietary factors? Should an arterial blood gas or further metabolic workup be considered, especially given stimulant use?

- Multiple medication discontinuations in 2026 due to side effects or ineffectiveness. Given the provisional ASD diagnosis, could atypical pharmacodynamics or sensory sensitivities be contributing to medication intolerance? Should future medication trials use lower starting doses or closer monitoring?

- Ferritin has remained borderline low (16 !' 21 ng/mL) for 5+ years, with declining iron saturation, despite normal hemoglobin. Given history of endometriosis, occult GI bleeding in 2023, and the known association between low ferritin and fatigue/cognitive symptoms in ADHD, should iron supplementation and source investigation be considered?

